

APSC 221 – ECONOMICS AND BUSINESS PRACTICES IN ENGINEERING

Course Syllabus – Summer 2026

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IMPORTANT COURSE INFORMATION

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory.
See: <http://www.queensu.ca/encyclopedia/t/traditional-territories>

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: <https://www.queensu.ca/inclusive/>

COURSE INSTRUCTOR

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E-mail: aref.s@queensu.ca

Office hour: By Appointment

APSC 221 (S K3)

COURSE DESCRIPTION

This course will provide students in the Engineering program with the ability to appropriately incorporate selected economic and business practices into the practice of engineering. The topics covered include introductions to: cost concepts, cost estimation, time value of money, cash flow analysis, the effects of taxes and inflation on project economics, replacement analysis, risk and change management, and financial strength and viability of a new business venture. Assignments and examples are based on situations from engineering-based industries.

(0/0/0/36/0) (Mathematics/Natural Sciences/Complementary Studies/Engineering Science/Engineering Design)

PREREQUISITE KNOWLEDGE

This course is designed for learners with no economics background; however, they will need to be comfortable with the foundations of algebra and with Microsoft Excel.

MODE OF INSTRUCTION

The course will be delivered through minimum a weekly 2h lecture – lecture hours to be determined at the end of week 1, and all sessions will be recorded and posted in OnQ for students who are unable to attend live.

COURSE LEARNING OUTCOMES (CLO)

By the end of this course, students should be able to:

CLO	Description	Indicator ¹
CLO 1	Recognize different cost concepts and apply them using a variety of cost estimation techniques.	EC – Economics
CLO 2	Solve cash flow analysis problems utilizing the time value of money.	EC – Economics
CLO 3	Determine the effect of taxes and inflation on project viability.	EC – Economics
CLO 4	Apply replacement analysis concepts to determine minimum equivalent annual costs.	EC – Economics
CLO 5	Examine risk and change management approaches for project management.	EC - Management
CLO 6	Assess the financial strength and viability of a new venture.	EC - Management
CLO 7	Write a basic business plan.	EC - Management

COURSE EVALUATION

Assessment Weighting

Assessment Tool	Due Date (before 23:59 PM ET)	Weight	Alignment with CLOs
Quizzes		20%	1-7
Q0: Course Introduction	Week 1 by May 10 th @ 11:59 PM	2%	N/A
Q1: Chapter 1, 2, 3	Week 2 by May 17 th @ 11:59 PM	2%	1,2
Q2: Chapter 4, 5	Week 3 by May 24 th @ 11:59 PM	2%	2
Q3: Chapter 6	Week 4 by May 31 st @ 11:59 PM	2%	2
Q4: Chapter 7, 8	Week 5 by June 7 th @ 11:59 PM	2%	3, 4
Q5: Chapter 9, 10	Week 6 by June 14 th @ 11:59 PM	2%	3, 5

¹ For further information on graduate attributes and indicators, please click [here](#).

Assessment Tool	Due Date (before 23:59 PM ET)	Weight	Alignment with CLOs
Q6: Chapter 11, Change Management	Week 8 by June 28 th @ 11:59 PM	2%	5
Q7: Chapter 12, 13	Week 9 by July 5 th @ 11:59 PM	2%	6
Q8: Chapter 14, 15	Week 10 by July 12 th @ 11:59 PM	2%	6, 7
Q9: Chapter 16, Circular Economy	Week 11 by July 19 th @ 11:59 PM	2%	7
Assignments	Day 7 of Weeks 5, 10, 11, 12	20%	1-7
Assignment 1	Week 5 by June 7 th @ 11:59 PM	5%	1-4
Assignment 2		15%	5, 6, 7
Part A	Week 10 by July 12 th @ 11:59 PM		
Part B	Week 11 by July 19 th @ 11:59 PM		
Part C	Week 12 by July 26 th @ 11:59 PM		
Midterm	Week 7 - TBD	20%	1-4
Final Exam	Please see the Final Exam schedule posted by OUR Exams Office	40%	1-7
		100%	

ASSESSMENT DESCRIPTIONS

Quizzes

There are 10 short quizzes in this course which align with the chapter readings. These quizzes are designed to provide learners with immediate feedback on their knowledge and are taken on the course website. You will have a seven-day window of opportunity to initiate a quiz.

Universal Design: It should take you about 30 minutes to complete each quiz once you start it, we have set up OnQ to give you 45 minutes for each quiz. Thus, unless you have an accommodation for more than double time, no extra time will be provided, and late submissions will not be accepted. You will have up to two attempts at each quiz. You do not have to attempt each quiz twice; however, please do. Your score will be the **highest** of both attempts.

Assignments

There are two assignments in this course. Assignment 1 is worth 5%. Assignment 2 is broken into Part A , B and C; the entire assignment is worth 15%. These assignments will be submitted to an assignment submission folder within onQ. All assignments should have a signed cover page attached. A 10% a day penalty will be deducted for any late submissions.

Midterm Exams

There is only one midterm for the course taking place in Week 7. Midterm will be only approximately 90 minutes in duration, subject to change, and will take place online using online proctoring. The midterm is closed-book assessment. The only permitted aid to use is a Casio 991 calculator.

Final Exam

The final exam is a closed-book exam. The final exam will cover all material from the course textbook and materials posted on the course webpage. Students must write their exam on the day and time scheduled by the University. You should not schedule vacations, travel, etc. during the exam period.

GRADING

All assessments on this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established [Grade Point Index](#).

Feedback on Assessments

The teaching team will provide feedback on graded activities. You can expect feedback on your assessments within two weeks of the due date.

Accessing Your Final Grade

Your final grades will be shown on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

COURSE MATERIALS

Required Textbook

Lecture Slides as well as any material posted in OnQ.

Other Material

All other course material is accessible via OnQ.

Required Calculator

A Casio 991 is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one term spanning 12 weeks. Learners can expect to invest on average 7-9 hours per week in this course. Learners who adhere to a pre-determined study schedule are more likely to successfully complete the course.

WEEKLY COURSE LEARNING OUTCOMES

Week	Learning Outcomes	Assessment
1	Course introduction	Quiz 0
2	By the end of this week, learners will be able to: • Cost Concepts and Design Economics [CLO1] • Cost-Estimation Techniques [CLO1] • Time Value of Money [CLO2]	Quiz 1 [CLO1][CLO2]
3	By the end of this week, learners will be able to: • Cash Flow Analysis [CLO2] • Present Worth, Annual Worth and Payback Methods [CLO2]	Quiz 2 [CLO2]
4	By the end of this week, learners will be able to: • Internal and External Rates of Return Methods [CLO2]	Quiz 3 [CLO2]
5	By the end of this week, learners will be able to: • Taxes [CLO3] • Replacement Decisions [CLO4]	Quiz 4 [CLO3][CLO4] Assignment 1 [CLO1] [CLO2] [CLO3] [CLO4]
6	By the end of this week, learners will be able to: • Inflation and Price Indices [CLO3] • Dealing with Uncertainty and Risk [CLO5]	Quiz 5 [CLO3][CLO5]
7		Midterm [CLO1][CLO2][CLO3][CLO4]
8	By the end of this week, learners will be able to: • Risk Management [CLO5] • Change Management [CLO5]	Quiz 6 [CLO5]

Week	Learning Outcomes	Assessment
9	By the end of this week, learners will be able to: - Recognizing Opportunities and Generating Ideas [CLO6] - Feasibility Analysis [CLO6]	Quiz 7 [CLO6]
10	By the end of this week, learners will be able to: - Assessing a New Venture's Financial Strength and Viability [CLO6] - Writing a Business Plan [CLO7]	Quiz 8 [CLO6][CLO7] Assignment 2 Part A [CLO5] [CLO6]
11	By the end of this week, learners will be able to: Basic management processes and concepts [CLO7]	Quiz 9 [CLO7] Assignment 2 Part B [CLO6]
12		Assignment 2 Part C [CLO7]
	TBD by OUR Exams Office	Final Exam [CLO1-7]

COURSE COMMUNICATION

QUESTIONS ABOUT COURSE MATERIAL

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

OFFICE HOURS

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

CONFIDENTIAL MATTERS

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 48 hours.

ABSENCES (ACADEMIC CONSIDERATIONS) AND MISSED ASSIGNMENTS

For information on academic considerations due to extenuating circumstances, please review the information on the [Smith Engineering website](#). Note that unacceptable reasons include extra-curricular activities, travel plans, scheduling conflicts, generally behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel is not an acceptable reason for granting academic considerations.

LATE SUBMISSION POLICY

In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (as described above). In the absence of an approved consideration request, the normal late penalty will apply as described in the assignment descriptions or any course/departamental policies. Once the submission folder or the quiz in OnQ closes, no further submissions will be accepted. Unless otherwise specified in the details of the assessment, a deduction of 10% per 24-hours will be applied after the date and time shown in OnQ.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

NETIQUETTE

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

Following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.

5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

STUDENT CODE OF CONDUCT

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.

<https://www.queensu.ca/nonacademicmisconduct/policies>

COPYRIGHT

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's **express consent**. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the [Professional Engineers Ontario Code of Ethics](#). Information on policies concerning academic integrity is available in the [Queen's University Code of Conduct](#), in the [Senate Academic Integrity Policy Statement](#), on the [Smith Engineering website](#), and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

PLAGIARISM/USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI) TOOLS:

Students must submit their own work and cite the work that is not theirs. Any other use constitutes a Departure from Academic Integrity.

Generative AI writing tools are permissible only if explicitly noted in the activity and/or assignment instructions. If any GenAI tools have been used, you must disclose the use and purpose and include proper citation of the generated materials. For information on how to cite the use of GenAI tools, please visit Queen's University Library.

INVALID EXAMS

An exam may be deemed invalid if it is disrupted; if instructions for a remote or online exam are not followed; if incorrect materials are uploaded; or if any circumstance arises that prevents the verification of academic integrity. When an exam is declared invalid, the situation will be reviewed, and the student may be granted an opportunity to re-write the exam based on the outcome of that review.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you including the [Library](#), [Student Academic Success Services \(Learning Strategies and Writing Centre\)](#), and [Career Services](#). Learners are encouraged to visit the Smith Engineering [Current Students](#) web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

ABSENCES (ACADEMIC CONSIDERATIONS) AND ACADEMIC ACCOMMODATIONS

For academic accommodations and considerations please review the information on the [Smith Engineering website](#).

ACCOMMODATIONS FOR DISABILITIES

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the QSAS website

(<https://www.queensu.ca/studentwellness/accessibility-services/>)

to learn more about academic accommodations. To start the registration process with QSAS, click the **Access Ventus** button found on the Ventus student portal:

<https://www.queensu.ca/studentwellness/accessibility-services/ventus>

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:

<https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students>

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

REQUESTS FOR RELIGIOUS AND OTHER HUMAN RIGHTS-BASED ACCOMMODATIONS IN SMITH ENGINEERING

RELIGIOUS ACCOMMODATIONS

Students in need of accommodation for religious observance are asked to complete the [Religious Observance and Academics Activities](#) form within a week of receiving their course syllabus (each term) and submit the form via email to the Engineering Program Advisor, (Accommodations & Considerations), engineering.aac@queensu.ca

- Alternative assignments are considered a “reasonable accommodation” under the Ontario Human Rights Code
- Please refer to the [Queen’s Multi-faith Calendar](#) for all of this year’s faith dates
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca

OTHER HUMAN-RIGHTS BASED ACCOMMODATIONS

Engineering students who need accommodations on Human Rights grounds are asked to contact, in confidence, the Engineering Program Advisor, (Accommodations & Considerations) engineering.aac@queensu.ca or to call 613-533-6000 x 78013.

TECHNICAL SUPPORT

Some basic comfort level with basic hardware and software skills are required for this course. If you require technical assistance, please contact [Technical Support](#).

SUPPORTIVE PERSONAL COUNSELLING

If at any time you find yourself feeling overwhelmed, anxious, sad, lonely, or distressed, consider confidential [personal counselling and wellness services](#) offered by Smith Engineering and the [Queen’s student wellness services](#).